

$$9) \ y = \frac{-x^2 + 2x + 15}{x + 4}; \quad [-3, 5]$$

$$10) \ y = \frac{x^2 + x - 6}{-x + 3}; \quad [-3, 2]$$

$$11) \ y = -2\sin(2x); \quad [-\pi, \pi]$$

$$12) \ y = \sin(2x); \quad [-\pi, \pi]$$

For each problem, determine if Rolle's Theorem can be applied. If it can, find all values of c that satisfy the theorem. If it cannot, explain why not.

$$13) \ y = \frac{x^2 - x - 12}{x + 4}; \quad [-3, 4]$$

$$14) \ y = \frac{-x^2 - 2x + 8}{-x + 3}; \quad [-4, 2]$$

$$15) \ y = \frac{-x^2 + 36}{x + 7}; \quad [-6, 6]$$

$$16) \ y = \frac{-x^2 + 4}{4x}; \quad [-2, 2]$$

$$17) \ y = 2\tan(x); \quad [-\pi, \pi]$$

$$18) \ y = -2\cos(2x); \quad [-\pi, \pi]$$